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## 1 1. Variable Separable Method

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The simplest and most straightforward technique for solving first-order Ordinary Differential Equations is the **Variable Separable** method. Whenever you are faced with a new DE, you should always check if it can be solved using this method before attempting more complex techniques.

### 1.1 1. Concept Overview

**Definition:** A first-order differential equation is considered “separable” if it can be written entirely in the form:

$$f(y)dy = g(x)dx$$

This means you can completely isolate all terms containing the dependent variable  $y$  (including  $dy$ ) on one side of the equals sign, and all terms containing the independent variable  $x$  (including  $dx$ ) on the other side.

### 1.2 2. Working Rule (Step-by-Step)

To solve a differential equation using the variable separable method, follow these steps:

1. **Rearrange the Equation:** Use algebraic manipulation (factoring, cross-multiplying, dividing) to get all  $y$ -terms with  $dy$  on the left side, and all  $x$ -terms with  $dx$  on the right side.
2. **Integrate Both Sides:** Apply the integral sign to both sides of the separated equation:  $\int f(y)dy = \int g(x)dx$ .
3. **Add the Constant:** After integrating, add a single arbitrary constant of integration (usually  $C$ ) to one side of the equation (conventionally the  $x$ -side).
4. **Apply Initial Conditions (If Given):** If the problem provides specific values for  $x$  and  $y$  (an Initial Value Problem), plug them into your general solution to solve for the exact value of  $C$ .

### 1.3 3. Solved Questions

Here are four questions from the lecture notes, demonstrating standard separation, factoring, initial value problems, and pre-substitution.

#### 1.3.1 Question 1: Standard Separation

**Solve:**  $(x + 1)\frac{dy}{dx} = x(y^2 + 1)$

**Solution:**

1. **Separate the variables:** Divide by  $(y^2 + 1)$  and divide by  $(x + 1)$  to move terms:

$$\frac{1}{y^2 + 1}dy = \frac{x}{x + 1}dx$$

2. **Manipulate the right side for easier integration:** Notice that  $\frac{x}{x+1} = \frac{x+1-1}{x+1} = 1 - \frac{1}{x+1}$

$$\frac{1}{y^2 + 1} dy = \left(1 - \frac{1}{x+1}\right) dx$$

3. **Integrate both sides:**

$$\int \frac{1}{y^2 + 1} dy = \int \left(1 - \frac{1}{x+1}\right) dx$$

$$\tan^{-1}(y) = x - \ln|x+1| + C$$

4. **Final General Solution:**

$$y = \tan(x - \ln|x+1| + C)$$


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### 1.3.2 Question 2: Factoring Before Separation

**Solve:**  $(xy^2 + x)dx + (yx^2 + y)dy = 0$

**Solution:**

1. **Factor out common terms:**

$$x(y^2 + 1)dx + y(x^2 + 1)dy = 0$$

2. **Move the  $dy$  term to the right side:**

$$x(y^2 + 1)dx = -y(x^2 + 1)dy$$

3. **Separate variables:** Divide by  $(x^2 + 1)$  and  $(y^2 + 1)$ :

$$\frac{x}{x^2 + 1} dx = -\frac{y}{y^2 + 1} dy$$

4. **Integrate both sides:** Multiply both sides by 2 to perfectly match the derivative of the denominators:

$$\frac{1}{2} \int \frac{2x}{x^2 + 1} dx = -\frac{1}{2} \int \frac{2y}{y^2 + 1} dy$$

$$\frac{1}{2} \ln|x^2 + 1| = -\frac{1}{2} \ln|y^2 + 1| + C$$

5. **Simplify using logarithm rules:**

$$\ln(x^2 + 1) + \ln(y^2 + 1) = 2C$$

$$\ln[(x^2 + 1)(y^2 + 1)] = C' \implies (x^2 + 1)(y^2 + 1) = e^{C'} = K$$

(Where  $K$  is just a new arbitrary constant).

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## 1.3.3 Question 3: Initial Value Problem (IVP)

**Solve:**  $x \frac{dy}{dx} + \cot y = 0$  given the initial condition  $y = \frac{\pi}{4}$  when  $x = \sqrt{2}$ .

**Solution:**

1. **Separate the variables:**

$$x \frac{dy}{dx} = -\cot y$$

$$\frac{1}{\cot y} dy = -\frac{1}{x} dx \implies \tan y dy = -\frac{1}{x} dx$$

2. **Integrate both sides:**

$$\int \tan y dy = -\int \frac{1}{x} dx$$

$$-\ln |\cos y| = -\ln |x| + \ln C$$

(Using  $\ln C$  instead of  $C$  makes simplification easier).

3. **Simplify the General Solution:** Multiply by  $-1$ :

$$\ln |\cos y| = \ln |x| - \ln C \implies \ln(\cos y) = \ln\left(\frac{x}{C}\right)$$

$$\cos y = \frac{x}{C} \implies x = C \cos y$$

4. **Apply Initial Conditions to find  $C$ :** Substitute  $x = \sqrt{2}$  and  $y = \pi/4$ :

$$\sqrt{2} = C \cos\left(\frac{\pi}{4}\right)$$

$$\sqrt{2} = C \left(\frac{1}{\sqrt{2}}\right) \implies C = \sqrt{2} \times \sqrt{2} = 2$$

5. **Final Particular Solution:**

$$x = 2 \cos y \quad \text{or} \quad y = \cos^{-1}\left(\frac{x}{2}\right)$$


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## 1.3.4 Question 4: Substitution Reducing to Separable

**Solve:**  $x^4 \frac{dy}{dx} + x^3 y = -\sec(xy)$

**Solution:**

1. **Factor out  $x^3$  on the left side:**

$$x^3 \left( x \frac{dy}{dx} + y \right) = -\sec(xy)$$

2. **Notice the product rule derivative:** The term  $(x \frac{dy}{dx} + y)$  is exactly the derivative of the product  $(xy)$ . Let's use a substitution.

- Let  $t = xy$
- Differentiate with respect to  $x$ :  $\frac{dt}{dx} = x \frac{dy}{dx} + y(1) = x \frac{dy}{dx} + y$

3. **Substitute  $t$  and  $dt/dx$  into the equation:**

$$x^3 \frac{dt}{dx} = -\sec t$$

4. **Now it is Variable Separable!**

$$\frac{1}{\sec t} dt = -\frac{1}{x^3} dx \implies \cos t dt = -x^{-3} dx$$

5. **Integrate:**

$$\int \cos t dt = -\int x^{-3} dx$$

$$\sin t = -\left(\frac{x^{-2}}{-2}\right) + C$$

$$\sin t = \frac{1}{2x^2} + C$$

6. **Resubstitute  $t = xy$  for the final solution:**

$$\sin(xy) = \frac{1}{2x^2} + C$$

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**Next Method:** What happens if the variables *cannot* be cleanly separated? Proceed to [Method 2: Homogeneous Differential Equations](#) to learn the next technique.